HQ-9 / HQ-9B Long-Range SAM — Battery Site Recognition Signature

Open-Source Site-Recognition Reference · curated register entry · compiled from open imagery-analysis literature

Scope. Recognition reference for the overhead (nadir) signature of a fielded HQ-9 / HQ-9B (CASIC; export designators HQ-9/P and HQ-9BE) long-range surface-to-air missile battery. This entry describes the *canonical site geometry* against which a candidate overhead observation may be compared. It is reference literature about the system — not an observation of any single site, and it carries no coordinate.

Canonical battery site signature. An HQ-9 / HQ-9B fire unit deploys to a prepared "petal" or "flower" pad: a roughly circular graded area, approximately 90–110 m in diameter, enclosed by an unpaved perimeter access road (~6 m width). Arranged radially around a central hub are approximately six (6) transporter-erector-launcher (TEL) revetments — open-topped earthen berms ~2–3 m high, oriented outward from the centre to give blast separation between launchers and the central engagement position. At the pad centre sits a raised circular hardstand, ~15–20 m in diameter, prepared for the engagement / fire-control radar (HT-233 or derivative). In dispersed layouts the engagement-radar hardstand may be offset from the launcher ring by roughly 180–380 m rather than centred.

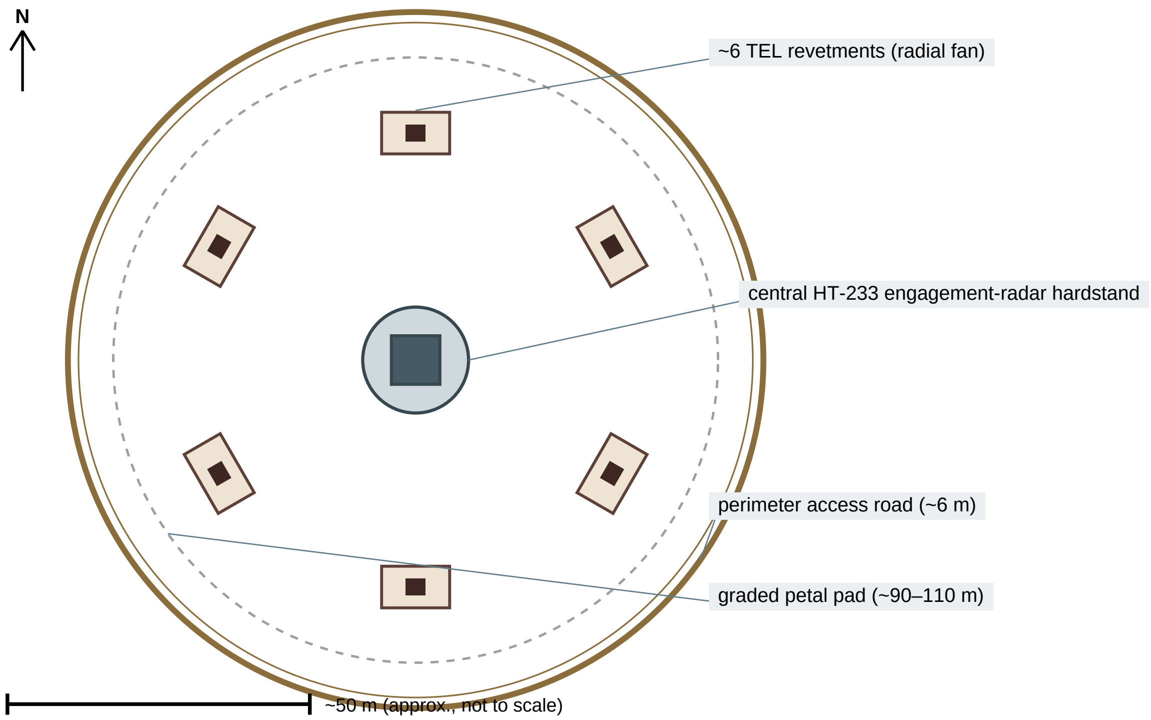

Figure 1. Schematic plan of the canonical HQ-9 / HQ-9B battery site signature (not to scale): circular petal pad enclosed by a perimeter access road; ~6 TEL revetments in a radial fan; central engagement-radar (HT-233) hardstand.

Diagnostic elements (descending reliability): (1) the radial / semi-radial arrangement of ~6 launcher revetments about a central hardstand — the primary signature; (2) individual canisterized TEL objects ~8–9 m in length (resolution-limited — a launcher count is often not reliably resolvable and should be reported as a *range*, never a fixed integer); (3) the central engagement-radar hardstand / berm; (4) the circular perimeter access road enclosing the pad.

Mobility platform. The HQ-9 / HQ-9/P TEL rides a heavy 8×8 special wheeled chassis of the Taian (Wanshan) WS-series lineage — the most consistently attributable deep-tier element of the system.

Recognition caveat. The petal / ring pad geometry is shared with several S-300 / S-400-family batteries and can be imitated by inflatable or radar-reflective decoy layouts. A single overhead pass showing this geometry is *consistent with* an HQ-9 / HQ-9B emplacement but is not, on its own, a positive identification: system-type confirmation requires a second, discipline-independent look (repeat imagery, an emitter fix, or a corroborating textual report). Treat a single-pass geometry match as a candidate signature only.

Compiled from open-source imagery-analysis literature (OSINT SAM-site site-analysis references and published air-defence battery-layout studies). Reference / recognition material — open distribution.